

ERIC JI

[GitHub](#) [Homepage](#) [LinkedIn](#)

PERSONAL INFORMATION

Citizenship: U.S. Citizen, **Phone:** +1 (518) 265-5649, **Email:** ericji3@illinois.edu

EDUCATION

University of Illinois Urbana-Champaign August 2024 – Present
Ph.D. in Electrical and Computer Engineering advised by Dr. Alexander Schwing GPA: 3.73/4.0

University of Illinois Urbana-Champaign August 2020 – May 2024
B.S. in Computer Engineering with Highest Honors GPA: 3.91/4.0

Relevant Courses: Computer Vision, Machine Learning, Pattern Recognition, Optimization, Random Processes

PUBLICATIONS

E. Ji, Q. Hu, W. Ma, S. Jain, Y. Li, M. N. Do, and Y. Liu, “AC3S: Adaptive Conditioning for 3D-Aware Synthetic Data Generation.” European Conference on Computer Vision (ECCV 2026).

E. Ji, B. Dong, B. Samanthula, and N. Zhou, “2D-FACT: Dual-Domain Fake Image Detection Against Text-to-Image Generative Models.” MIT Undergraduate Research Technology Conference (URTC 2023).

S. K. Kamtikar, **E. Ji**, N. K. Uppalapati, G. Krishnan, and G. Chowdhary, “Realistic Simulation Environments to Achieve Visual Servoing on Soft Continuum Arms in Constrained Environments.” Fourth International Workshop on Machine Learning for Cyber-Agricultural Systems (MLCAS 2022).

RESEARCH EXPERIENCE

Schwing Research Group May 2026 – Present
Advised by Dr. Alexander Schwing

- Exploring techniques to improve human action understanding in egocentric videos through generative modeling

Computational Imaging Group August 2024 – May 2026
Advised by Dr. Minh N. Do in collaboration with Dr. Yaoyao Liu

- Developed a CAD-to-image feature matching pipeline for a vision-based electronics manufacturing inspection system
- Improved pose control and realism of objects generated by diffusion models to aid in synthetic data generation

Vision Group August 2023 – May 2024
Advised by Dr. Svetlana Lazebnik

- Leveraged image forensics techniques to develop a classifier capable of detecting synthetically generated images
- Analyzed the impact of various strategies for integrating Fourier-domain signals into deep neural networks

WORK EXPERIENCE

Honda Research Institute May 2026 – August 2026

- Developed a state-of-the-art method for error detection in long-form videos of procedural tasks
- Derived theoretical foundations and insights supporting the use of diffusion models for classification

Siemens Digital Industries Software May 2025 – August 2025

- Implemented a diffusion-based synthetic data generation pipeline to aid in foundation model pretraining
- Optimized multimodal foundation model inference for deployment in industrial applications

TEACHING

Teaching Assistant – ECE 484: Principles of Safe Autonomy

TECHNICAL SKILLS

Languages C, C++, Java, Python
Tools Blender, Git, NumPy, OpenCV, Pandas, PyTorch, Scikit-Learn